

On Closed-Form Preintegration for a Class of Mixed-Invariant Systems in $SE_n(3)$

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Abstract—A large class of nonlinear systems in engineering and robotics evolves on geometric manifolds, such as fixed-wing aircraft, quadrotors, etc., whose kinematic motion can be described using Lie groups, in particular $SE_n(3)$. The existing techniques involve numerical integration methods such as Runge-Kutta integration to propagate their motion forward in time. However, these methods are approximate solutions and do not respect the geometric constraints of the nonlinear differential equation, leading to higher computational costs and numerical integration errors. To this end, we propose a geometric closed-form approach to solve the initial value problem for a class of nonlinear systems that evolves on the $SE_n(3)$ Lie group. Through numerical simulations, we show that our closed-form solution is more efficient than the Runge-Kutta 4th-order (RK-4) integrator and reduces floating-point operations by 45% in a strapdown-inertial-navigation (SINS) case study. We believe that the efficiency and accuracy gains of our approach warrant the general adoption of this method for numerical integration in applications such as SINS.

Index Terms—Numerical algorithms, aerospace, algebraic/geometric methods.

I. INTRODUCTION

MANY problems in science, engineering, and robotics require integration schemes, such as strapdown-inertial-navigation-systems (SINSs) [1], [2], and simultaneous localization and mapping (SLAM) [3]. These dynamics can be highly nonlinear and extremely sensitive to small perturbations, resulting in the explosion of the solution due to numerical errors. Furthermore, computing the integration can be prohibitively expensive for resource-constrained systems such as micro-aerial vehicles, which require integration for inertial navigation, trajectory tracking, etc. Thus, an efficient method that computes the integration with lower

computational cost, higher precision and superior numerical stability under perturbation is essential.

A large class of problems in control theory and robotics involves controlling and estimating the motion of a rigid body in space, such as uncrewed aircraft control, etc. In recent years, a class of nonlinear systems, also known as mixed-invariant systems [4], [5], particularly those that evolve on $SE_n(3)$ manifolds [6], [7], [8], has been widely adopted to model, control and estimate the motion of various robotic systems including aerial, wheeled, legged robots [9], and robotic manipulators. Furthermore, the $SE_n(3)$ representation is also utilized to model the SLAM problem, which involves not only the motion of the robot but also the dynamics of the landmarks with respect to the robot.

In the robotics literature, IMU preintegration has been extensively studied [10], [11], [12], [13], for its application in SINS and SLAM. The authors in [14] proposed a three-step procedure, i.e., lift-solve-retract, for the preintegration, where lifting and retracting refer to the matrix exponentials of the left and right group action, respectively. Barrau and Bonnabel [13] extended this notion of preintegration using the group-theoretic approach. Both of these approaches resorted to numerical integration for the preintegration. Most recently, the authors in [8] have developed a closed-form solution for the preintegration. However, the results are limited to $SE_2(3)$ and cannot be trivially extended to the general $SE_n(3)$ group.

To overcome these challenges, in this letter, we develop a novel formulation for calculating the closed-form preintegration approach by modeling the system's evolution as a matrix differential equation. We take advantage of the unique representation of the matrix to obtain a closed-form integration solution for a class of mixed-invariant systems [4], [5] (composed of vector fields of left and right invariant vector fields) evolving in the $SE_n(3)$ manifold. We show that an exact solution to the initial value problem can be obtained by exploiting this matrix structure. Since our method is exact, the time step has no impact on the accuracy of the solution. In addition, computational complexity is also reduced by about 45% compared to a Runge-Kutta solver of 4th order (RK-4) solver. Finally, we present a case study of SINS to validate the effectiveness of our proposed method.

The remainder of this letter is organized as follows. In Section II, we provide the mathematical preliminaries related

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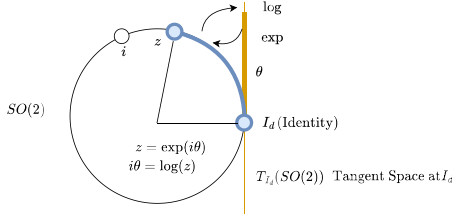


Fig. 1. Illustration of Lie group and its corresponding Lie algebra for $SO(2)$, adopted from [17].

to the representation of mixed invariant systems. In Section III, we formulate the geometric integration problem for mixed-invariant systems evolving on $SE_n(3)$ manifold. In Section IV, we provide our closed-form solution for geometric integration leveraging the special structure of mixed-invariant systems. In Section V, we compare our approach with the numerical RK-4 integration method for the SINS case study. Section VI summarizes this letter.

Notations: In this letter, I and $\mathbf{0}$ denote the identity and zero matrices, respectively, of appropriate dimensions. A vector is denoted by x and M is a matrix. $GL(n, \mathbb{R})$ is the set of $n \times n$ invertible matrices, together with the operation of matrix multiplication. $\mathcal{G}$ denotes a Lie group and $\mathfrak{g}$ denotes its corresponding Lie algebra.

II. MATHEMATICAL PRELIMINARIES

This section introduces the definition of mixed-invariant systems. Furthermore, we describe the $SE_n(3)$ Lie group and its associated Lie algebra. A detailed description of matrix Lie groups can be found in [15], [16].

Definition 1: A mixed-invariant system [4], [5] evolving on a Lie group $\mathcal{G}$ is a system with the differential equation:

$$\dot{X} = MX + XN \quad (1)$$

where $X \in \mathcal{G} \subset GL(n, \mathbb{R})$, and M , and N can be any constant matrices, satisfying: $Ad_{X^{-1}}M + N \in \mathfrak{g}$, where $\mathfrak{g}$ is the associated Lie algebra of $\mathcal{G}$.

Intuitively, for every Lie group, its corresponding Lie algebra lies in the tangent space at the identity element and there exist a bijection map (locally near the identity element) from the Lie group to its Lie algebra using $\log$ and $\exp$, respectively, as shown in Fig. 1. We extend the notion of the adjoint action of the Lie group for a mixed-invariant system, where $M, N \notin \mathfrak{g}$, but satisfies the following criterion: $Ad_{X^{-1}}M + N \in \mathfrak{g}$.

Definition 2: For every $X \in \mathcal{G}$, the adjoint operator for a mixed-invariant system, $Ad_X : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}^{n \times n}$, and it is defined as:

$$Ad_X M = XMX^{-1} \quad (2)$$

We now present the definition of the group of n direct spatial isometries $SE_n(3)$ [6], [7], [8], which is a specific instantiation of the mixed-invariant system. It is defined as follows:

$$X := \begin{bmatrix} R & p_1 & \dots & p_n \\ 0 & 1 & \dots & 0 \\ \vdots & 0 & \ddots & 0 \\ 0 & 0 & \dots & 1 \end{bmatrix} := \begin{bmatrix} R & P \\ \mathbf{0} & I_n \end{bmatrix} \quad (3)$$

where $R \in SO(3)$ denotes the attitude of the body with respect to the world frame, $p_i \in \mathbb{R}^3 \forall i \in \{1, \dots, n\}$ denotes the translational states in the world frame. We write this as a block matrix for ease of manipulation later, where $P := [p_1 \dots p_n] \in \mathbb{R}^{3 \times n}$.

The corresponding $\mathfrak{se}_n(3)$ Lie algebra can be represented as:

$$[x]^\wedge := \begin{bmatrix} [\omega]^\wedge & u_1 & \dots & u_n \\ 0 & 0 & \dots & 0 \\ \vdots & 0 & \ddots & 0 \\ 0 & 0 & \dots & 0 \end{bmatrix} := \begin{bmatrix} \Omega & U \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \quad (4)$$

where $x = [u_1 \dots u_n \omega]^\top \in \mathbb{R}^{3(n+1)}$, and $[\cdot]^\wedge$ denotes the wedge operator that maps elements from $\mathbb{R}^m$ to a corresponding Lie algebra with dimension m , denoted as,

$$[\omega]^\wedge = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix}. \quad (5)$$

$\omega \in \mathbb{R}^3$ denotes the angular velocity in the body frame, $\Omega := [\omega]^\wedge \in \mathfrak{so}(3)$ is the corresponding skew-symmetric matrix corresponding to ω , and $u_i \in \mathbb{R}^3 \forall i \in \{1, \dots, n\}$ usually denotes the translational input in the body frame. We write this as a block matrix for ease of manipulation later, where $U := [u_1 \dots u_n] \in \mathbb{R}^{3 \times n}$. These definitions will play an important role later when we derive the closed-form expression for the initial value problem for the class of systems that evolves on the $SE_n(3)$ manifold.

III. PROBLEM FORMULATION

Consider a system evolving on a specific class of mixed-invariant vector field in $SE_n(3)$ Lie group:

$$\dot{X} = MX + XN \quad (6)$$

where $X \in SE_n(3) \subset \mathbb{R}^{(n+3) \times (n+3)}$ denotes the system state, M and $N \in \mathbb{R}^{(n+3) \times (n+3)}$ are the input matrices. The matrices M and N have the following form:

$$W = \begin{bmatrix} \Omega_W & A \\ \mathbf{0} & B \end{bmatrix}, \quad W \in \{M, N\}, \quad (7)$$

where $\Omega_W \in \mathfrak{so}(3) \subset \mathbb{R}^{3 \times 3}$, $A \in \mathbb{R}^{3 \times n}$ and $B \in \mathbb{R}^{n \times n}$ is a nilpotent matrix with index 2 (which implies that $B^k = \mathbf{0} \forall k \geq 2$). We will later show that the matrices B_M and B_N also satisfy the following relation: $B_M = -B_N$. Note that our approach is applicable for constant matrices M and N , and thus, requires piecewise constant angular velocities ω .

Remark 1: A large class of robotic systems can be modeled using a mixed-invariant vector field [18] defined in (6). For example, SINS, SLAM, etc.

Given an initial state X_0 , the problem is to find a closed form solution to the initial value problem for the dynamics defined in (1). In most existing works, the above initial value problem is approximated by numerical integration methods, such as Euler integration, Runge-Kutta integration, etc. As mentioned earlier, the main issue with approximate methods is that they are susceptible to numerical errors and depend on the numerical precision of the computational hardware. To address this, in this letter, we derive the closed-form expression of the

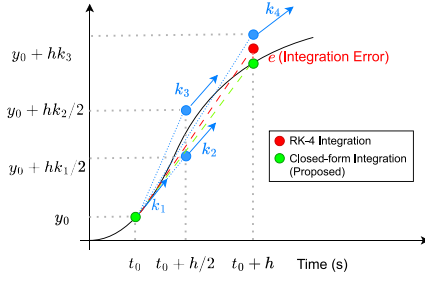


Fig. 2. Illustration of numerical integration RK-4 [19] and our proposed geometric integration method.

initial value problem for a class of nonlinear systems defined by a mixed-invariant system. The main idea is to leverage the specific structure of the evolution of the matrix differential equation that allows us to transform the series-form expression of its solution to a closed form.

Figure 2 depicts our approach in which each sample is a discrete point where integration is performed. The initial state is represented by the last sample between each sample points, and the inputs are kept constant while integrating. Note that this matches the implementation of SINS using IMU measurements, where the system state is computed by integrating the system dynamics on each IMU update.

IV. CLOSED-FORM MATRIX EXPONENTIAL FOR $SE_n(3)$

In this section, we derive the closed-form expression for the integration of a mixed-invariant system in $SE_n(3)$ Lie group. Note that achieving a closed-form solution is tractable, since B is a nilpotent matrix. We will leverage this unique $SE_n(3)$ structure in the derivation of Theorem 1. Before we present our main results, we present the following Lemmas 1 and 2, which will play an important role in the derivation of our main results. Lemma 1 allows us to identify the unique structure in the matrix representation of the mixed invariant system in $SE_n(3)$ and Lemma 2 provides a matrix exponential form solution for the initial value problem.

Lemma 1: Consider the system defined in (6). Suppose B_M and B_N represent the matrices corresponding to the bottom right block of size $n \times n$ of M and N , respectively. Then,

$$B_M + B_N = 0 \quad (8)$$

Proof: Consider the block representation of matrices M and N as follows:

$$M = \begin{bmatrix} \Omega_M & A_M \\ \mathbf{0} & B_M \end{bmatrix} \quad (9)$$

$$N = \begin{bmatrix} \Omega_N & A_N \\ \mathbf{0} & B_N \end{bmatrix} \quad (10)$$

where $\Omega_M, \Omega_N \in \mathfrak{so}(3) \subset \mathbb{R}^{3 \times 3}$, $A_M, A_N \in \mathbb{R}^{3 \times n}$ and $B_M, B_N \in \mathbb{R}^{n \times n}$. As a consequence of the definition of the mixed-invariant system Definition 1, the following criterion must be satisfied: $Ad_{X^{-1}}M + N \in \mathfrak{g}$.

Using the above criterion, we establish the following matrix relationship:

$$Ad_{X^{-1}}M + N = X^{-1}MX + N \quad (11)$$

$$\begin{aligned} Ad_{X^{-1}}M + N &= \begin{bmatrix} R^T & -R^T P \\ \mathbf{0} & I_n \end{bmatrix} \begin{bmatrix} \Omega_M & A_M \\ \mathbf{0} & B_M \end{bmatrix} \begin{bmatrix} R & P \\ \mathbf{0} & I_n \end{bmatrix} \\ &\quad + \begin{bmatrix} \Omega_N & A_N \\ \mathbf{0} & B_N \end{bmatrix} \\ &= \begin{bmatrix} R^T \Omega_M R + \Omega_N R^T (\Omega_M P + A_M - P B_M) + A_N & \\ \mathbf{0} & B_M + B_N \end{bmatrix} \end{aligned} \quad (12)$$

The resulting matrix in (12) must be an element of the Lie algebra $\mathfrak{g}$ and have the same form as (4). Thus, we obtain $B_M + B_N = 0$, which proves the result. ■

Lemma 2: For the matrix differential equation defined as $\dot{X} = MX + XN$. The solution to the initial value problem can be described as

$$X_t = e^{Mt} X_0 e^{Nt} \quad (13)$$

where X_0 is the initial value of the states.

Proof: Suppose X_t has the form $X_t = e^{Mt} X_0 e^{Nt}$, which can be also represented as $X_t = e^{Mt} X_0 (e^{-Nt})^{-1}$. Using the chain rule and the product rule for computing derivatives, we can obtain the derivative as follows,

$$\begin{aligned} \frac{d}{dt} X_t &= M e^{Mt} X_0 (e^{-Nt})^{-1} + e^{Mt} X_0 e^{Nt} N e^{-Nt} e^{Nt} \\ &= M e^{Mt} X_0 (e^{-Nt})^{-1} + e^{Mt} X_0 e^{Nt} N \\ &= M X_t + X_t N \end{aligned} \quad (14)$$

This proves the result. ■

Lemma 2 provides us with the form of the solution of the matrix differential equation (6). However, the main challenge is that the matrix exponentials e^{Mt} and e^{Nt} generally do not have a closed-form solution. We now present our main results, where we derive the closed-form expression for the initial value problem in Theorem 1.

Theorem 1: Consider a mixed-invariant system in $SE_n(3)$ defined in (6), with $M \in \mathbb{R}^{(n+3) \times (n+3)}$ and $N \in \mathbb{R}^{(n+3) \times (n+3)}$ matrices satisfying $Ad_{X^{-1}}M + N \in \mathfrak{g}$. If the sub-matrix B defined in (7) is nilpotent with index 2, then the initial value problem for the matrix differential equation (1) with $X_0 = \begin{bmatrix} R_0 & P_0 \\ \mathbf{0} & I_n \end{bmatrix}$ as the initial condition has a closed-form expression. Moreover, the closed form expression has the following form:

$$X(t) = \begin{bmatrix} R_M R_0 R_N & R_M R_0 P_N + (R_M P_0 + P_M)(I + Bt) \\ \mathbf{0} & I_n \end{bmatrix} \quad (15)$$

where

$$R_M := e^{\Omega_M t} R_N := e^{\Omega_N t},$$

$$P_M := P(\Omega_M t, A_M t, -Bt) P_N := P(\Omega_N t, A_N t, Bt),$$

$$P(\Omega_W, A, B) = A + AB/2 + \Omega_W(c_1(\theta)I_n + c_2(\theta)B) + \Omega_W^2 A(c_2(\theta)I_n + c_3(\theta)B),$$

$$\theta := \|\omega\|, c_1(\theta) := \frac{1 - \theta^2/2 - \cos(\theta)}{\theta^2}.$$

The coefficients $c_1(\cdot)$, $c_2(\cdot)$ and $c_3(\cdot)$ are defined as $c_1(\theta) := \frac{1 - \theta^2/2 - \cos(\theta)}{\theta^2}$, $c_2(\theta) := \frac{\theta - \sin \theta}{\theta^3}$ and $c_3(\theta) := \frac{\theta^2/2 - \theta^4/24 + \cos \theta - 1}{\theta^4}$.

Proof: We want to find the closed-form expression for $X(t) = e^{Mt}X(0)e^{Nt}$, which we have shown in Lemma 2 is the solution to the initial value problem for the matrix differential equation defined by (6). Note that M and N are not elements of the $\mathfrak{se}_n(3)$; however, matrices M and N can be represented as follows:

$$W = \begin{bmatrix} \Omega_W A \\ \mathbf{0} & B \end{bmatrix}, \quad (16)$$

where W is a dummy variable representing either M or N , $\Omega_W = \Omega_M$ (or Ω_N) depending on the specific instantiation of W as either M or N .

We begin with the series expansion of the exponential of e^W , which can be expressed as follows:

$$e^W = \sum_{k=0}^{\infty} \frac{W^k}{k!}. \quad (17)$$

Now, we find the powers of W . Since B is nilpotent with index 2, higher powers are zero matrices when power is greater or equal to 2. The square of W can be expressed as:

$$W^2 = \begin{bmatrix} \Omega_W^2 A & \Omega_W A + AB \\ \mathbf{0} & B^2 \end{bmatrix}. \quad (18)$$

For $k > 1$, the power of W can be written as:

$$W^k = \begin{bmatrix} \Omega_W^k A & \Omega_W^{k-1} A + \Omega_W^{k-2} AB \\ \mathbf{0} & B^k \end{bmatrix} \quad (19)$$

This allows us to write the series form of the matrix exponential of e^W as:

$$e^W = \sum_{k=0}^{\infty} \frac{W^k}{k!} = \begin{bmatrix} \sum_{k=0}^{\infty} \frac{\Omega_W^k A}{k!} & P(\Omega_W, A, B) \\ \mathbf{0} & \sum_{k=0}^{\infty} \frac{B^k}{k!} \end{bmatrix} \quad (20)$$

where $P(\Omega_W, A, B) = A + \sum_{k=2}^{\infty} \left[\frac{\Omega_W^{k-1} A}{k!} + \frac{\Omega_W^{k-2} AB}{k!} \right]$.

We now transform the infinite sum as a summation from 0 to ∞ , as follows:

$$P(\Omega_W, A, B) = A + \sum_{k=0}^{\infty} \left[\frac{\Omega_W^{k+1} A}{(k+2)!} + \frac{\Omega_W^k AB}{(k+2)!} \right] \quad (21)$$

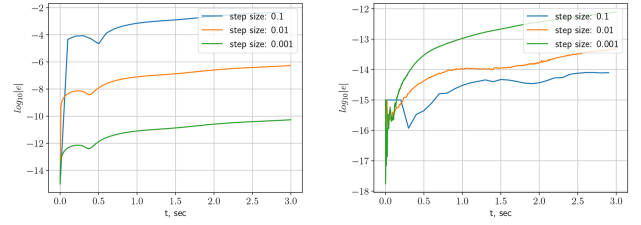
In order to further simplify the expression in (21), we split it into the summation of even and odd terms as follows,

$$P(\Omega_W, A, B) = A + \sum_{k=0}^{\infty} \left[\frac{\Omega_W^{2k+1} A}{(2k+2)!} + \frac{\Omega_W^{2k} AB}{(2k+2)!} \right] \quad (22)$$

$$+ \sum_{k=0}^{\infty} \left[\frac{\Omega_W^{2k+2} A}{(2k+3)!} + \frac{\Omega_W^{2k+1} AB}{(2k+3)!} \right] \quad (23)$$

Define $\theta := \|\omega\|$. Splitting the terms into odd and even series summation allows us to replace the series expression into $\sin(\cdot)$ and $\cos(\cdot)$ terms as follows:

$$\begin{aligned} \sum_{k=0}^{\infty} \frac{\Omega_W^{2k+1} A}{(2k+2)!} &= \sum_{k=0}^{\infty} \frac{(-1)^k \theta^{2k} \Omega_W A}{(2k+2)!} \\ &= \frac{1 - \theta^2/2 - \cos(\theta)}{\theta^2} \Omega_W A \end{aligned} \quad (24)$$



(a) Log error between RK-4 integration and reference trajectory with time. (b) Log error between our geometric integration and reference trajectory with time.

Fig. 3. Comparison of numerical errors between Runge-Kutta-4th order with our proposed geometric integration approach for various step sizes.

$$\begin{aligned} \sum_{k=0}^{\infty} \frac{\Omega_W^{2k+2} A}{(2k+3)!} &= \sum_{k=0}^{\infty} \frac{(-1)^k \theta^{2k} \Omega_W^2 A}{(2k+3)!} = \frac{\theta - \sin \theta}{\theta^3} \Omega_W^2 A \quad (25) \\ \sum_{k=0}^{\infty} \frac{\Omega_W^{2k+1} AB}{(2k+3)!} &= \sum_{k=0}^{\infty} \frac{(-1)^k \theta^{2k} \Omega_W AB}{(2k+3)!} = \frac{\theta - \sin \theta}{\theta^3} \Omega_W AB \end{aligned} \quad (26)$$

$$\begin{aligned} \sum_{k=0}^{\infty} \frac{\Omega_W^{2k} AB}{(2k+2)!} &= \frac{AB}{2} + \sum_{k=1}^{\infty} \frac{\Omega_W^{2k} AB}{(2k+2)!} \\ &= \frac{AB}{2} + \sum_{k=0}^{\infty} \frac{\Omega_W^{2k+2} AB}{(2k+4)!} \\ &= \frac{AB}{2} + \sum_{k=0}^{\infty} \frac{(-1)^k \theta^{2k} \Omega_W^2 AB}{(2k+4)!} \\ &= \frac{AB}{2} + \frac{\theta^2/2 - \theta^4/24 + \cos \theta - 1}{\theta^4} \Omega_W^2 AB \end{aligned} \quad (27)$$

As a consequence of the substitutions carried out in (24), (25), (27) and (27), we get the closed-form expression for $P(\Omega_W, A, B)$ as follows:

$$\begin{aligned} P(\Omega_W, A, B) &= A + AB/2 + \Omega_W A (c_1(\theta) I_n + c_2(\theta) B) \\ &\quad + \Omega_W^2 A (c_2(\theta) I_n + c_3(\theta) B) \end{aligned} \quad (28)$$

where $c_1(\theta) := \frac{1 - \theta^2/2 - \cos(\theta)}{\theta^2}$, $c_2(\theta) := \frac{\theta - \sin \theta}{\theta^3}$ and $c_3(\theta) := \frac{\theta^2/2 - \theta^4/24 + \cos \theta - 1}{\theta^4}$. The complete matrix exponential of e^{Mt} (or e^{Nt}) is described as:

$$\begin{aligned} e^{Mt} &= \begin{bmatrix} e^{\Omega_M t} P(\Omega_M t, A_M t, B_M t) \\ \mathbf{0} & I_n + B_M t \end{bmatrix} \\ &= \begin{bmatrix} R_M & P_M \\ \mathbf{0} & I_n + B_M t \end{bmatrix} \end{aligned} \quad (29)$$

Using the result from Lemma 1, we know $B_M = -B_N$. The closed-form expression for the matrices R_M and R_N can be computed using Rodrigues' formula [8]. Plugging (29) into (13), we obtain the closed-form solution of $X(t)$:

$$\begin{aligned} X(t) &= e^{Mt} X_0 e^{Nt} \\ &= \begin{bmatrix} R_M & P_M \\ \mathbf{0} & I_n + B_M t \end{bmatrix} \begin{bmatrix} R_0 & P_0 \\ \mathbf{0} & I_n \end{bmatrix} \begin{bmatrix} R_N & P_N \\ \mathbf{0} & I_n + B_N t \end{bmatrix} \\ &= \begin{bmatrix} R_M R_0 R_N & R_M R_0 P_N + (R_M P_0 + P_M)(I + B_N t) \\ \mathbf{0} & I_n \end{bmatrix} \end{aligned}$$

This proves the result. ■

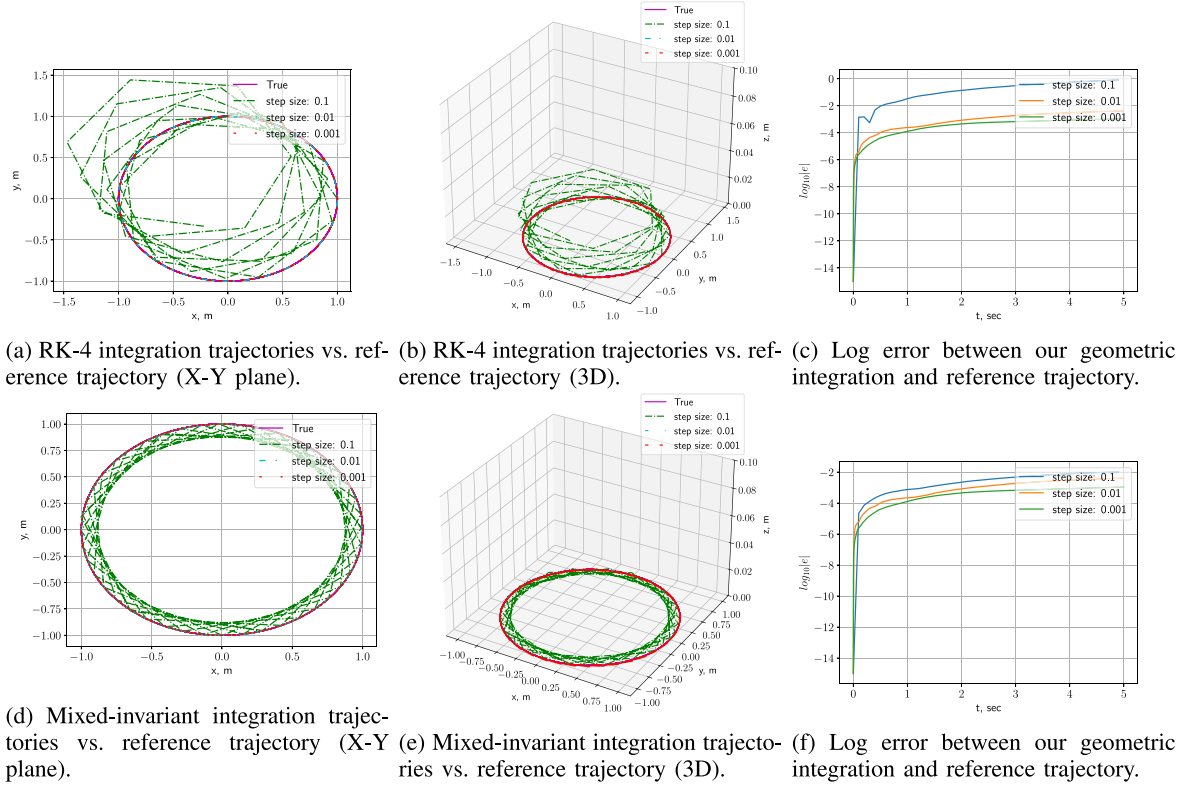


Fig. 4. Comparison of numerical errors between Runge-Kutta-4th order with our proposed geometric integration approach for different step sizes under input perturbation.

V. CASE STUDY: STRAPDOWN-INERTIAL NAVIGATION

In this section, we evaluate the effectiveness of our proposed method by comparing it with Runge-Kutta 4th-order (RK-4) integration, which is the most popular approach for numerical integration for many engineering applications [19]. All experiments were carried out using a Dell Precision-3581 workstation laptop with a 13th Gen Intel Core i7 processor. We consider the initial value problem for SINS of a UAV moving a circular path at a constant velocity as a case study for our evaluation. We first present the differential equation describing the UAV's motion using the mixed-invariant representation in the $SE_2(3)$ manifold.

A. $SE_2(3)$ Lie Group Embedded Kinematics

The SINS propagation requires solving the initial value problem with kinematics given by:

$$\dot{R} = R[\omega]^\wedge \dot{v} = Ra + a_g \dot{p} = v \quad (30)$$

where $p \in \mathbb{R}^3$ denotes the position in the world frame, $v \in \mathbb{R}^3$ denotes the translational velocity in the world frame, $a = [a_x \ a_y \ a_z]^\top \in \mathbb{R}^3$ represents the translational acceleration in the body frame, and $a_g = [0 \ 0 \ g]^\top \in \mathbb{R}^3$ represents the fixed gravitational acceleration in the world frame. Note that our approach can be generalized to any constant body acceleration a and any constant world acceleration a_g . The kinematics model in (30) can be embedded in the $SE_2(3)$ Lie group:

$$\dot{X} = MX + XN, \quad (31)$$

where $X = \begin{bmatrix} R & v & p \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \in SE_2(3)$ represents the state, L and $R \in \mathbb{R}^{5 \times 5}$ represent the input of the SINS. We will leverage the special structure of the M and N matrices for SINS where $M := \begin{bmatrix} 0 & A_m \\ 0 & -B_n \end{bmatrix}$, $N := \begin{bmatrix} \Omega_W & A_n \\ 0 & B_n \end{bmatrix}$, $A_m := [a_g \ 0]$, $A_n := [a \ 0]$, and $B_n := \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$.

B. Numerical Simulations

We consider an aggressive UAV orbit maneuver moving at a velocity of 5 m/s around a circular trajectory of radius 1 m with the body x -axis pointing in the direction of the velocity for 3 seconds. In order to compare the different integration approaches, we first sample the simulated gyroscope and accelerometer readings at the rate at which the integration is performed. We first compare the number of floating point operations between RK-4 and our proposed method by constructing equation graphs using CasADI [20] for both methods used to perform integration (see [20] for more details on equation graphs). We can clearly observe a 45% reduction in the number of floating point operations per propagation step as shown in Table I. Note that this is only for a single integration step. If the step size is increased further, the number of floating-point operations for RK-4 will increase significantly compared to our approach, which is unaffected due to its closed-form nature. For example, in most numerical solvers such as CasADI, a relative tolerance is set for the RK-4 integration scheme, which adapts the step size based on the tolerance. In such cases, the computational costs grow at the rate of $\mathcal{O}(n)$, where n is the additional

TABLE I
COMPARISON OF THE NUMBER OF FLOATING OPERATIONS BETWEEN
RK-4 AND OUR PROPOSED METHOD PER INTEGRATION STEP

Integration Scheme	Floating point operations/ Integration step
RK-4	677
Mixed-invariant	360

number of steps compared to our proposed method where it is constant.

In Fig. 3, we first compare the error between RK-4 and our proposed geometric integration approach with varying step sizes. Figure 3(a) shows that as the step size is reduced, the error in RK-4 is reduced and the integration errors are significantly reduced. However, reducing the step size requires the Runge-Kutta method to perform more computations than our closed-form method. In contrast, it is evident from Fig. 3(b) that our closed-form method has errors near the numerical precision of floating-point operations, showing that it is independent of the step size of the sampling.

To showcase another important benefit of our proposed approach, we perturb the input measurements, i.e., IMU measurements, in the SINS problem to show how our closed-form approach improves the stability of the integration solution over the RK-4 method. It is well known that in the presence of a disturbance in the IMU measurements, the SINS performs a random walk behavior as a result of the accumulation of errors [21]. We perturb the IMU measurements by a factor of 10^{-4} at every step. Figure 4 shows the comparison between RK-4 and our geometric approach when the input measurements are perturbed. It can be observed from Figs. 4(c) and 4(f), the errors in our approach are 2-orders less in magnitude compared to the RK-4 integration approach. This highlights a key advantage of our proposed method and its future adoption in the autopilot community. Our source code is open and available for download <https://github.com/CogniPilot/cyecca>.

VI. CONCLUSION

In this letter, we presented a closed-form geometrical integration method for a class of nonlinear systems evolving on Lie groups, i.e. mixed-invariant systems. Through numerical simulations, we show that our approach requires 45% fewer floating point operations to compute the numerical integration using the industry standard Runge-Kutta 4th-order (RK-4) integration. We showed that our proposed method is more stable than RK-4 in the presence of perturbation in the system dynamics as a result of noise and disturbances. In the future, we will integrate this method to solve the SINS and SLAM problems and improve their run-time performance, especially for resource-constrained systems.

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